MAVIA Data Tables

This document explains the main SQLite tables used by MAVIA. Default Django tables such as auth_user, django_session, django_migrations, and django_content_type are framework support tables.

Course and Outline Tables

Table	Purpose	Important Fields
lessons_coursegroup	Stores the course itself, such as Life Science or Properties of Matter.	title, description, created_at, updated_at
lessons_courseoutline	Stores the uploaded course outline PDF connected to one course.	course_id, outline_file, is_approved, uploaded_at, approved_at
lessons_outlinenode	Stores the parsed outline hierarchy: modules, topics, and subtopics.	course_id, parent_id, title, order, depth
lessons_outlineedge	Stores prerequisite or relationship edges between outline nodes.	course_id, source_id, target_id, score, validation_status

PDF and Learning Content Tables

Table	Purpose	Important Fields
lessons_learningmaterial	Stores each uploaded lesson PDF, extracted raw text, generated JSON, mapped outline node, status, and errors.	course_id, outline_node_id, module_node_id, title, pdf_file, extracted_text, generated_json, status
lessons_learningobject	Stores the final learner-facing content blocks used for audio and questions.	material_id, kind, title, content, image_prompt, order
lessons_learningobjectprerequisiteedge	Stores the learner-path graph connections between learning objects in a module.	course_id, module_node_id, source_id, target_id, relation_type, score, validation_status
lessons_question	Older/internal lesson question table if used by the lessons app.	learning material or lesson references, question text and answer fields depending on migration

Concept Graph Tables

Table	Purpose	Important Fields
lessons_extractedconcept	Stores concepts extracted from lesson materials or outline content.	title, description, type/difficulty/order fields depending on migration
lessons_conceptsource	Stores where an extracted concept came from, such as a specific material or block.	concept_id, source references, excerpt/source metadata
lessons_conceptprerequisiteedge	Stores prerequisite relationships between extracted concepts.	source_id, target_id, score, explanation, validation_status
lessons_moduleconceptdagstate	Stores the status/state of the module concept graph generation.	course/module references, status, generated data/error metadata

Question Generation Tables

Table	Purpose	Important Fields
question_generation_generatedquestion	Stores the actual generated questions for each learning object.	node_id, question_text, question_format, choices, correct_answer, explanation, bloom_level, difficulty, category, intended_difficulty, difficulty_match
question_generation_learnerresponse	Stores learner answers to generated questions.	learner_id, question_id, selected_answer, is_correct, answered_at
question_generation_generationrun	Stores one backend question-generation process triggered by the teacher.	material_id, node_id, status, started_at, finished_at
question_generation_generationevent	Stores trace logs/events for a generation run, such as classifier_loading, node_started, question_generated, saved, or error.	run_id, seq, event_type, message, data, created_at

Most Important Tables For Your Defense

lessons_learningmaterial is the main PDF table. It stores the uploaded PDF path, extracted text, and generated_json. The generated_json contains classified_blocks, learning_objects, learning_objectives, assessments, teacher_notes, ignored_blocks, image_descriptions, and lesson_playlist.

lessons_learningobject stores the final learner-facing learning objects as separate rows. Audio and question generation use these learning objects.

question_generation_generatedquestion stores the actual questions generated from each learning object.

question_generation_generationrun and question_generation_generationevent are backend tracing tables. They help MAVIA show progress and diagnose errors, but the teacher mainly cares about the generated question list.

Simple Explanation

Course structure is stored in CourseGroup, CourseOutline, and OutlineNode. Uploaded PDFs and their extraction results are stored in LearningMaterial. Final learning objects are stored in LearningObject. Learner-path connections are stored in LearningObjectPrerequisiteEdge. Questions are stored in GeneratedQuestion, while GenerationRun and GenerationEvent keep backend progress logs.